

Regular expressions (regex)

Thursday, June 18, 2026

i Today: Thursday June 18

Before class

- **Reading Quiz** (Canvas): [Data Computing Ch. 17](#) §17.1–17.2 — Regular expressions

Plan for today

- **Project oral check-in #1** runs during today's work session (signup on Canvas)
- What a **regex** is, and the three jobs it does: **detect**, **replace**, **extract**
- Building blocks: literals, `.`, character classes `[ ]`, alternation `|`
- **Quantifiers** `?` `*` `+` `{n}`, **anchors** `^` `$`, and **escaping**
- **Capturing** with `( )` and pulling pieces out
- In-class activity

Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [DC Ch. 17](#) · [stringr cheatsheet](#) · [regex101.com](#)

What is a regex?

Patterns in strings

A **regular expression** (regex) is a compact way to describe a **pattern** in text. Lots of everyday strings follow patterns:

Kind	Examples
Phone numbers	(651) 696-6000 · +1 651.696.6000
Times	10:30 AM · 13:15.54

Kind	Examples
Durations	1:59:40 (Kipchoge's sub-2-hour marathon)
US postal codes	16801 · 55105-1362
Stray whitespace	" henry "

A regex lets you say “match anything shaped *like this*” without listing every case.

Three jobs regex does

Across all of data cleaning, regex does three things:

Job	base R	stringr (tidyverse)	Often with
Detect a pattern	<code>grepl()</code>	<code>str_detect()</code>	<code>filter()</code>
Replace a pattern	<code>gsub()</code>	<code>str_replace_all()</code>	<code>mutate()</code>
Extract a match	<code>regmatches()</code>	<code>str_extract()</code>	<code>mutate()</code>

The textbook uses base R (`grepl`, `gsub`); the stringr versions do the same jobs with tidier names. We'll show both.

`grepl()`: does the pattern appear?

`grepl(pattern, x)` returns TRUE/FALSE for each string. The **pattern is the quoted string**. Here: names containing “shine” (`ignore.case = TRUE` ignores capitalization):

```
NameList |>
  filter(grepl("shine", name, ignore.case = TRUE)) |>
  head(5)
```

```
# A tibble: 5 x 3
  name      sex  total
<chr>    <chr> <int>
1 Sunshine F      4959
2 Shineka  F       150
3 Shine    F        95
4 Shine    M        44
5 Sunshine M        37
```

`str_detect(name, regex("shine", ignore_case = TRUE))` does the same thing.

Regex building blocks

Literals & special characters

The simplest patterns are **literal**: "able" matches any string containing “able” — “capable”, “tabled” — but is **case-sensitive** by default (“Able” won’t match unless `ignore.case = TRUE`).

These characters are **special** — they mean something other than themselves:

```
. ^ $ [ ] { } | ( ) + * ?
```

For example `.` means “any single character”, so `".ble"` matches “able”, “soluble”, “fungible”, “Bible”.

Character classes []

Square brackets mean “**any one** of these characters”:

- `[aeiou]` → any vowel
- `[^aeiou]` → **not** a vowel (the `^` *inside* brackets means “except”) → a consonant
- `[0-9]` → any digit; `[a-z]` → any lowercase letter

```
NameList |>
  filter(grepl("mn", name, ignore.case = TRUE)) |> # literal "mn"
  head(3)
```

```
# A tibble: 3 x 3
  name    sex    total
<chr> <chr> <int>
1 Autumn F      104408
2 Sumner M        2287
3 Amna   F        1099
```

Tip

A vertical bar `|` means “**or**”: `"rain|snow|sleet|hail"` matches any of those four words.

Putting characters in a row

Two patterns side by side means “this **then** that”. `^[^aeiou].[^aeiou].[^aeiou]` reads as:

- `^` start of string, then
- `[^aeiou]` a consonant, `.` anything, `[^aeiou]` a consonant, `.` anything, `[^aeiou]` a consonant

i.e. the 1st, 3rd, and 5th letters are consonants:

```
NameList |>
  filter(grepl("^[^aeiou].[^aeiou].[^aeiou]", name, ignore.case = TRUE)) |>
  head(3)
```

```
# A tibble: 3 x 3
  name    sex    total
  <chr>  <chr>   <int>
1 James  M       5091189
2 Robert M       4789776
3 David  M       3565229
```

Quantifiers

How many times?

A quantifier follows a pattern and says **how many times** it repeats:

Quantifier	Means
<code>?</code>	zero or one time
<code>*</code>	zero or more times
<code>+</code>	one or more times
<code>{n}</code>	exactly n times
<code>{n,m}</code>	between n and m times
<code>{n,}</code>	n or more times

Warning

The *Data Computing* reading swaps the definitions of `+` and `*`. **In R (and every real regex engine):** `*` = zero-or-more, `+` = one-or-more. Trust the table above and the code below — not the book on this one point.

Quantifiers in action

```
x <- c("Mr.", "Ms", "Mrs.", "Miss", "Man", "Miocene")
x[grepl("M[aeiou]+", x)] # M then ONE or more vowels
```

```
[1] "Miss"      "Man"       "Miocene"
```

```
x[grepl("M[aeiou]*", x)] # M then ZERO or more vowels (matches Mr.)
```

```
[1] "Mr."      "Ms"       "Mrs."     "Miss"     "Man"      "Miocene"
```

Three or more vowels in a row, [aeiou]{3,}:

```
NameList |>
  filter(grepl("[aeiou]{3,}", name, ignore.case = TRUE)) |>
  head(3)
```

```
# A tibble: 3 x 3
  name    sex    total
  <chr>  <chr>  <int>
1 Louis  M      389910
2 Louise F      331551
3 Isaiah M      177412
```

Anchors & escaping

Start ^ and end \$

Outside of square brackets, ^ and \$ are **anchors**:

- ^ = the **start** of the string
- \$ = the **end** of the string

Names that **end** in a vowel, [aeiou]\$:

```
NameList |>
  filter(grepl("[aeiou]$", name)) |>
  group_by(sex) |>
  summarise(total = sum(total), .groups = "drop")
```

```
# A tibble: 2 x 2
  sex      total
<chr>   <int>
1 F       96702371
2 M       21054791
```

Girls' names end in a vowel almost five times as often as boys' names.

Escaping special characters

To match a special character **literally**, **escape** it with `\\`. So `\\.` means a real period, and `\\$` a real dollar sign (not the end-of-string anchor).

```
currency <- c("$100.95", "45¢")
gsub("^\\$|¢$", "", currency) # strip a leading $ or trailing ¢
```

```
[1] "100.95" "45"
```

Here `^\\$` is a literal `$` at the start; `¢$` is a literal `¢` at the end. Without the `\\`, that first `$` would mean “end of string”.

Capturing & extracting

Parentheses capture

Wrap part of a pattern in `( )` to **capture** it — to pull that piece out. `str_extract()` returns the matched text; with capture groups, `gsub()` can refer back to group 1 as `\\1`.

Names with **four vowels in a row**, extracting those vowels:

```
BabyNames |>
  group_by(name) |>
  summarise(total = sum(count), .groups = "drop") |>
  mutate(match = str_extract(name, "[aeiouAEIOU]{4}")) |>
  filter(!is.na(match)) |>
  arrange(desc(total)) |>
  head(4)
```

```
# A tibble: 4 x 3
  name      total match
  <chr>    <int> <chr>
1 Louie   29293 ouie
2 Sequoia 3275 uoia
3 Gioia    480 ioia
4 Keiaira   81 eiai
```

Backreferences with \\1

The regex `".*([aeiou])$"` = “anything, then a captured final vowel”. In `gsub()`, `replacement = "\\1"` keeps **only** the captured group — so the whole name collapses to its last vowel:

```
re <- ".*([aeiou])$"
NameList |>
  filter(grepl(re, name)) |>
  mutate(vowel = gsub(re, "\\1", name)) |>
  group_by(sex, vowel) |>
  summarise(total = sum(total), .groups = "drop") |>
  pivot_wider(names_from = sex, values_from = total)
```

```
# A tibble: 5 x 3
  vowel      F      M
  <chr>    <int> <int>
1 a      56088501 1844041
2 e      36432218 14341114
3 i       3693024  753311
4 o       403120  4041190
5 u        85508   75135
```

Names ending in “a”, “e”, “i” skew female; “o” skews male.

Cleaning with regex

Stripping junk out of numbers

Scraped numbers arrive with \$, ,, %, units. A character class lists everything to delete at once:

```
debt <- tibble(
  country = c("World", "United States*", "Japan"),
  debt     = c("$56,308 billion", "$17,607 billion", "$9,872 billion"),
  percGDP  = c("64%", "73.60%", "214.30%")
)
debt |>
  mutate(
    debt     = gsub("[$,%]|billion", "", debt),
    percGDP  = gsub("[, %]", "", percGDP)
  )
```

```
# A tibble: 3 x 3
  country      debt      percGDP
  <chr>        <chr>    <chr>
1 World      "56308 " 64
2 United States* "17607 " 73.60
3 Japan      "9872 " 214.30
```

`[$,%]` deletes any \$, ,, or %; `|billion` also deletes the word “billion”.

Trimming whitespace

`^ +| +$` matches one-or-more spaces at the **start** *or* the **end**:

```
s <- "  My name is Julia  "
gsub("^ +| +$", "", s)
```

```
[1] "My name is Julia"
```

Tip

stringr has friendly shortcuts for these: `str_remove_all()`, `str_trim()` (ends), and `str_squish()` (ends + collapse internal runs). Same regex ideas underneath.

Activity

In-class activity

Open `day23-activity.R` in RStudio and work through the parts in order. (Today's session also hosts **oral check-in #1** — work on the activity when it isn't your turn.)

- **Part 1:** detect patterns with `grepl()` / `str_detect()` + `filter()` on `BabyNames`.
- **Part 2:** build patterns from **character classes**, **quantifiers**, and **anchors** (including a “Caitlin”-style name and a phone-number regex).
- **Part 3 (challenge):** **extract** with capture groups, and **clean** messy strings with `gsub()`.

[Activity file](#)

Wrap-up & looking ahead

What we covered

- A regex describes a **pattern**; it does three jobs: **detect** / **replace** / **extract**
- . any char · `[aeiou]` class · `[^aeiou]` negation · `|` or
- Quantifiers `?` `*` `+` `{n}` `{n,m}` `{n,}` (R: `+` = one-or-more, `*` = zero-or-more)
- Anchors `^` start, `$` end; **escape** specials with `\\`
- Capture with `( )`; pull pieces with `str_extract()` or `\\1` in `gsub()`
- Clean numbers/whitespace with `gsub()` (or `str_remove_all()`/`str_squish`)

Upcoming

- **Fri Jun 19: No class — Juneteenth**
- **Quiz 5** rescheduled to **Mon Jun 22**
- Keep building your **project**; oral check-in #1 today

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

[Syllabus](#) · [AI policy](#) · [Schedule](#) · [DC Ch. 17](#) · regex101.com